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Semiconductor structure and method for manufacturing same

Abstract

Provided is a semiconductor structure and a method for manufacturing the same. The method includes forming a spin on hard mask layer on a base, active areas arranged at intervals in the base, bit lines arranged at intervals and extending in a first direction on the base, each bit line electrically connected to at least one active area, and the spin on hard mask layer filled between the bit lines and covering the bit lines; removing part of the spin on hard mask layer to form first trenches arranged at intervals and extending in a second direction; forming first sacrificial layers in the first trenches; removing the spin on hard mask layer between the first sacrificial layers to form second trenches; forming first supporting layers in the second trenches; removing the first sacrificial layers and elongating the first trenches located between adjacent bit lines to the active areas.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8846540	12/2013	Lee	N/A	N/A
9425200	12/2015	Hwang	N/A	N/A
10411014	12/2018	Hwang	N/A	N/A
10586709	12/2019	Ko	N/A	N/A
11120992	12/2020	Huh	N/A	N/A
11296088	12/2021	Hwang	N/A	N/A
2009/0061615	12/2008	Cho	N/A	N/A
2014/0057442	12/2013	Lee et al.	N/A	N/A
2015/0126013	12/2014	Hwang et al.	N/A	N/A
2016/0329337	12/2015	Hwang et al.	N/A	N/A
2019/0172717	12/2018	Ko et al.	N/A	N/A
2019/0348418	12/2018	Hwang et al.	N/A	N/A
2021/0124258	12/2020	Ahn	N/A	H01L 21/0274
2021/0143012	12/2020	Huh	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101378034	12/2008	CN	N/A
109872967	12/2018	CN	N/A
112786437	12/2020	CN	N/A
113097144	12/2020	CN	N/A
113517233	12/2020	CN	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Application No. PCT/CN2022/070400, filed on Jan. 5, 2022, which claims priority to Chinese Patent Application No. 202111311525.3, filed on Nov. 8, 2021. The disclosures of International Application No. PCT/CN2022/070400 and Chinese Patent Application No. 202111311525.3 are hereby incorporated by reference in their entireties.

BACKGROUND

(1) With the development of semiconductor technologies, the application of a semiconductor structure is more and more extensive. A dynamic random access memory (DRAM) has gradually become a semiconductor memory device commonly used in an electronic device. The dynamic random access memory includes a plurality of memory cells, and each of which includes a transistor and a capacitor. The capacitor is used for storing data information and the transistor is used for controlling the reading or writing of the data information in the capacitor. The gate of the transistor is electrically connected to a word line (WL), and the transistor is controlled to be turned on and off by the voltage applied on the word line. One of the source and the drain of the transistor is electrically connected to a bit line (BL), while the other one of the source and the drain is connected to the capacitor, and the data information is stored or outputted via the bit line.

(2) In the related art, bit lines arranged at intervals and extending in a first direction are usually first formed on a base, and then first supporting layers are formed between adjacent bit lines, and the bit lines and the first supporting layers define filling holes. However, in the process of forming the filling holes, relatively more portion of each bit line far away from the base loses, which affects the performance of the bit line.

SUMMARY

(3) The disclosure relates to the technical field of semiconductors, and in particular to a semiconductor structure and a method for manufacturing the same.

(4) Embodiments of the disclosure provide a semiconductor structure and a method for manufacturing the same.

(5) A first aspect of the embodiments of the disclosure provides a method for manufacturing a semiconductor structure, which includes: forming a spin on hard mask layer on a base, in which a plurality of active areas arranged at intervals are provided in the base, a plurality of bit lines arranged at intervals and extending in a first direction are provided on the base, each of the bit lines is electrically connected to at least one of the active areas, and the spin on hard mask layer is filled between the bit lines and covers the bit lines; removing part of the spin on hard mask layer to form a plurality of first trenches arranged at intervals and extending in a second direction; forming first sacrificial layers in the first trenches, in which the first sacrificial layers are filled in the first trenches; removing the spin on hard mask layer between the first sacrificial layers to form second trenches; forming first supporting layers in the second trenches, in which the first supporting layers are filled in the second trenches; and removing the first sacrificial layers and elongating the first trenches located between adjacent bit lines to the active areas.

(6) A second aspect of the embodiments of the disclosure provides a semiconductor structure, which is obtained by the above-mentioned method, and thus has at least the advantage that the loss of a bit line is less.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a flowchart of a method for manufacturing a semiconductor structure in an

embodiment of the disclosure;
(2) FIG. **2** is a schematic structural diagram of a base and bit lines in an embodiment of the disclosure;
(3) FIG. **3** is a three-dimensional diagram after a photoresist layer is formed in an embodiment of the disclosure;
(4) FIG. **4** is a cross-sectional schematic diagram at A-A of FIG. **3**;
(5) FIG. **5** is a three-dimensional diagram after first sacrificial layers are formed in an embodiment of the disclosure;
(6) FIG. **6** is a cross-sectional schematic diagram at B-B of FIG. **5**;
(7) FIG. **7** is a three-dimensional diagram after second trenches are formed in an embodiment of the disclosure;
(8) FIG. **8** is a three-dimensional diagram after first supporting layers are formed in an embodiment of the disclosure;
(9) FIG. **9** is a three-dimensional diagram after retained spin on hard mask layer is removed in an embodiment of the disclosure;
(10) FIG. **10** is a cross-sectional schematic diagram at C-C of FIG. **9**;
(11) FIG. **11** is a schematic structural diagram after contact holes are elongated to active areas in an embodiment of the disclosure;
(12) FIG. **12** is a schematic structural diagram after first grooves are formed in an embodiment of the disclosure;
(13) FIG. **13** is a schematic structural diagram after a first filling layer is formed in an embodiment of the disclosure;
(14) FIG. **14** is a schematic structural diagram after a second sacrificial layer is exposed in an embodiment of the disclosure;
(15) FIG. **15** is a schematic structural diagram after first etching grooves are formed in an embodiment of the disclosure;
(16) FIG. **16** is a schematic structural diagram after a silicon-containing anti-reflection layer is penetrated with first etching grooves in an embodiment of the disclosure;
(17) FIG. **17** is a schematic structural diagram after first trenches are formed in an embodiment of the disclosure;
(18) FIG. **18** is a schematic structural diagram after first intermediate grooves are formed in an embodiment of the disclosure;
(19) FIG. **19** is a schematic structural diagram after a third sacrificial layer is formed in an embodiment of the disclosure;
(20) FIG. **20** is a schematic structural diagram after part of a third sacrificial layer is removed in an embodiment of the disclosure;
(21) FIG. **21** is a schematic structural diagram after first sacrificial layers are formed in an embodiment of the disclosure;
(22) FIG. **22** is a schematic structural diagram after layers on a spin on hard mask layer are removed in an embodiment the disclosure;
(23) FIG. **23** is a schematic structural diagram after a first conductive layer is formed in an embodiment of the disclosure;
(24) FIG. **24** is a schematic structural diagram after conductive pillars are formed in an embodiment of the disclosure; and
(25) FIG. **25** is a schematic structural diagram after a first protective layer is formed in an embodiment of the disclosure.

DETAILED DESCRIPTION

(26) In order to reduce the loss of a bit line during manufacturing a semiconductor structure, embodiments of the disclosure provide a method for manufacturing a semiconductor structure, in which a spin on hard mask layer filled between bit lines and covering the bit lines is formed, and by

utilizing a high selectivity ratio between the spin on hard mask layer and the bit lines, the loss of a portion of a bit line is reduced during subsequent etching, thereby ensuring the performance of the bit line.
(27) In order to make the above-mentioned objectives, features and advantages of embodiments of this disclosure more apparent and easy to be understood, the technical solutions in the embodiments of the disclosure will be clearly and completely described with reference to the drawings of the embodiments of this disclosure. Apparently, the described embodiments are only a part of embodiments of this application, not all of them. Based on the embodiments in this disclosure, any other embodiments obtained by an ordinary person skilled in the art without involving creative efforts are within the protection scope of the application.
(28) Referring to FIG. **1**, embodiments of the disclosure provide a method for manufacturing a semiconductor structure, which at least includes the following operations.
(29) At S**101**, a spin on hard mask layer is formed on a base. A plurality of active areas arranged at intervals are provided in the base, a plurality of bit lines arranged at intervals and extending in a first direction are provided on the base, each of the bit lines is electrically connected to at least one of the active areas, and the spin on hard mask layer is filled between the bit lines and covers the bit lines.
(30) Referring to FIG. **2** to FIG. **4**, the filling patterns shown in the drawings of the embodiments of the disclosure are only used to distinguish different structures in the drawings, and not used to indicate materials of the structures in the drawings. As shown in FIG. **2** to FIG. **4**, a base **100** is used for supporting layers formed on the base **100**, for example, a bit line **150** and a spin on hard mask layer **200**.
(31) As shown in FIG. **2** to FIG. **4**, a plurality of active areas **111** arranged at intervals are provided in the base **100**. The active areas **111** can be defined by shallow trench isolations (STIs) **112**. Specifically, part of the base **100** is removed to a preset depth by an etching process to form grooves surrounding a plurality of active areas **111**, and then an insulating material is deposited in the grooves to isolate the active areas **111** from each other. The insulating material may be silicon oxide, silicon nitride or the like.
(32) Exemplarily, the base **100** may include a substrate **110**, an insulating layer **120** and a barrier layer **130** stacked in sequence. Herein, the substrate **110** may be a semiconductor substrate **110**, for example, a silicon substrate, a germanium substrate, a silicon-germanium substrate, a germanium-arsenic substrate, a silicon-on-insulator (SOI) substrate or a germanium-on-insulator (GOI) substrate, etc. The substrate **110** may be doped or undoped, and exemplarily, the substrate **110** may be an N-type substrate or a P-type substrate.
(33) The above said active areas **111** are formed in the substrate **110** and exposed at the top surface of the substrate **110**. A plurality of word lines **113** arranged at intervals are also formed in the substrate **110**. As shown in FIG. **4**, a plurality of word lines **113** extend in a second direction and the word lines **113** are respectively insulated from the active areas **111**. Herein, the word lines **113** may be buried word lines (BWLs), and the active areas **111** are arranged obliquely to the extending direction of the word lines **113** so as to increase the arrangement density of the active area **111**.
(34) Still referring to FIG. **2** and FIG. **3**, the insulating layer **120** is formed on the substrate **110** and covers the active areas **111** to isolate and protect the active areas **111**. The material of the insulating layer **120** may be the same as the insulating material in the shallow trench isolations **112**. After the shallow trench isolations **112** are formed by depositing the insulating material, the insulating material is continuously deposited to form the insulating layer **120**, so as to simplify the manufacturing operations of the semiconductor structure.
(35) A barrier layer **130** is formed on the insulating layer **120**, and the barrier layer **130** corresponds to bit lines **150**. The material of the barrier layer **130** may be silicon nitride or silicon oxynitride, and the barrier layer **130** can be used as an etching stop layer in a subsequent process to reduce the etching of the insulating layer **120**. A plurality of contact holes are formed in the barrier layer **130**

and the insulating layer **120**, and the contact holes respectively expose the active areas **111**. Exemplarily, two contact holes correspond to one active area **111**, and each of the two contact holes exposes one end of the active area **111**.

(36) Still referring to FIG. **2** and FIG. **3**, a plurality of bit lines **150** arranged at intervals are provided on the base **100**. Each bit line **150** extends in a first direction, and is electrically connected to at least one active area **111**. Exemplarily, a bit line plug (also known as bit line contact) **140** is provided in each of the contact holes, and the bit line plug **140** is in contact with one of the active areas **111**. The bit line plugs **140** may be a plurality of columnar structures filled in the contact holes. Alternatively, as shown in FIG. **2**, the bit line plugs **140** may have a comb-teeth structure, in which each tooth of the comb is filled in each contact hole. When forming the bit line plugs **140** and the bit lines **150** by etching, the barrier layer **130** is also etched, such that the retained barrier layer **130** corresponds to each bit line plug **140**, i.e. the bit line plugs **140** located outside the contact holes are respectively deposited on the barrier layer **130**. It can be understood that, the cross-section at A-A shown in FIG. **4** is the plane located between two adjacent bit lines, and there is no barrier layer **130** in this plane.

(37) Still referring to FIG. **2** and FIG. **3**, in some possible examples, a bit line **150** includes a second conductive layer **151** and a second supporting layer **152** covering the second conductive layer **151**. The second conductive layer **151** extends in the first direction and is in contact with a bit line plug **140**, and the bit line **150** is electrically connected to an active area **111** via the bit line plug **140**. As shown in FIG. **2** and FIG. **3**, the second supporting layer **152** also covers the base **100** between two adjacent ones of the second conductive layers **151**.

(38) As shown in FIG. **2**, in the second supporting layer **152**, an oxide layer **153** is provided at either side of the second conductive layer **151**. For example, two oxide layers **153** are provided at two sides of the second conductive layer **151**, respectively. The oxide layers **153** are not in contact with the second conductive layer **151**. No oxide Layer **153** is shown in FIG. **3**.

(39) Herein, as shown in FIG. **2**, the material of the bit line plug **140** may be polysilicon. The second conductive layer **151** may be a metal layer or a metal stack. For example, the second conductive layer **151** includes a titanium nitride layer in contact with the bit line plug **140** and a tungsten layer located on the titanium nitride layer. The second supporting layer **152** may be a nitride layer, such as a silicon nitride layer. In the direction away from a sidewall of the second conductive layer **151**, there are a nitride, an oxide and another nitride (NON) in sequence.

(40) Referring to FIG. **3** and FIG. **4**, a spin on hard mask (SOH) layer **200** is filled between the bit lines **150** and covers the bit lines **150**. The selectivity ratio between the spin on hard mask layer **200** and the second supporting layer **152** of the bit line **150** is large, so that the etching loss of the second supporting layer **152** is less in the subsequent process for etching the spin on hard mask layer **200**, thus reducing the loss of the bit line **150** and ensuring the performance of the bit line **150**. Exemplarily, the selection ratio between the spin on hard mask layer **200** and the second supporting layer **152** is greater than or equal to 5.

(41) At S**102**, part of the spin on hard mask layer is removed to form a plurality of first trenches arranged at intervals and extending in a second direction.

(42) Referring to FIG. **4**, part of a spin on hard mask layer **200** is removed by a process of dry etching or wet etching to form a plurality of first trenches arranged at intervals and extending in a second direction. There is an angle between the second direction and the first direction, for example, the second direction is perpendicular to the first direction. After the first trenches are formed, the spin on hard mask layer **200** is divided into a plurality of pieces by the first trenches.

(43) It can be understood that, each of the first trenches includes a second groove located on each bit line and extending in the second direction, and filling holes located between two adjacent ones of the bit lines and in communication with the second groove. That is, in the process of forming the first trenches, the second grooves are formed in the spin on hard mask layer **200** located on the bit lines, and the filling holes are formed in the spin on hard mask layer **200** between the bit lines. The

second grooves are in communication with the filling holes located below the second grooves.
(44) At S**103**, first sacrificial layers are formed in the first trenches, in which the first sacrificial layers are filled in the first trenches.
(45) Referring to FIG. **5** and FIG. **6**, first sacrificial layers **220** are deposited in the first trenches, in which the first sacrificial layers fill up the first trenches. Exemplarily, the first sacrificial layers may be formed by a process of chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD) or the like. The material of the first sacrificial layers **220** can be an oxide, such as silicon oxide, to facilitate subsequent removal.
(46) At S**104**, the spin on hard mask layer between the first sacrificial layers is removed to form second trenches.
(47) Referring to FIG. **7**, the spin on hard mask layer **200** between the first sacrificial layers **220** is removed by etching after the first sacrificial layers **220** are formed. It can be understood that, the spin on hard mask layer **200** is removed in two operations, in which the first trenches are formed after removing part of the spin on hard mask layer **200** in the first operation, and second trenches **230** are formed after removing the retained spin on hard mask layer **200** in the second operation. The shape of each second trench **230** is roughly the same as the shape of each first trench.
(48) At S**105**, first supporting layers are formed in the second trenches, in which each of the first supporting layers is filled in the second trenches.
(49) Referring to FIG. **7** and FIG. **8**, first supporting layers **240** are deposited in the second trenches **230**, in which the first supporting layers **240** respectively fill up the second trenches **230**. The material of the first supporting layers **240** may be the same as the material of the second supporting layers **152** of the bit lines **150**, both of which are insulating materials, for example, silicon nitride.
(50) At S**106**, the first sacrificial layers are removed and the first trenches located between adjacent bit lines are elongated to the active areas.
(51) Referring to FIG. **8** to FIG. **11**, the first sacrificial layers **220** are removed by etching. Exemplarily, the first sacrificial layers **220** are removed by a process of wet etching with an acidic etchant, and there is a large selectivity ratio between the first sacrificial layers **220** and the first supporting layers **240**, so as to reduce damage of the first supporting layers **240**. After the first sacrificial layers **220** are removed, the first trenches **210** are exposed. As shown in FIG. **11**, the first trenches **210** located between adjacent bit lines **150** are also elongated to the active areas **111** to expose the active areas **11** in the first trenches **210**.
(52) In an possible embodiment, a base **100** includes a substrate **110**, an insulating layer **120** provided on the substrate **110**, a barrier layer **130** provided on the insulating layer **120** in sequence, in which a plurality of active areas **111** are provided in the substrate **110**, and the insulating layer **120** covers the active areas **111**. Accordingly, removing the first sacrificial layers **220** and elongating the first trenches **210** located between adjacent bit lines **150** to the active areas **111** includes the following operations.
(53) Referring to FIG. **8** to FIG. **10**, the first sacrificial layers **220** are etched to expose the first trenches **210**. Each of the first trenches **210** includes a second groove **211** located on the bit lines **150** and extending in the second direction, and filling holes **212** located between adjacent bit lines **150** and in communication with the second groove **211**. As shown in FIG. **9** and FIG. **10**, the second supporting layers **152** extending in the second direction and the first supporting layers **240** extending in the first direction define the aforementioned filling holes **212**.
(54) Referring to FIG. **11**, the insulating layer **120** is etched along the filling holes **212** of the first trenches **210**, so as to expose the active areas **111** by the filling holes **212**. As shown in FIG. **11**, a filling hole **212** penetrates through the insulating layer **120** and extends to an active area **111**, so that the active area **111** is exposed in the filling hole **212**, which facilitates the electrical connection between the active area **111** and a conductive pillar **250** formed in the filling hole **212**. It should be noted that, when a second supporting layer **152** also covers the base **100**, the filling hole **212** also penetrates through the second supporting layer **152**, that is, the filling hole **212** penetrates through

the second supporting layer **152** and the insulating layer **120**, and extends to the active area **111**.
(55) To sum up, according to the method for manufacturing a semiconductor structure in the embodiments of the disclosure, the spin on hard mask layer **200** filled between the bit lines **150** and covering the bit lines **150** is formed, and the high selectivity ratio between the spin on hard mask layer **200** and the bit lines **150** is utilized, thus the loss of portion of each bit line **150** far away from the base **100** during subsequent etching is reduced, thereby ensuring the performance of the bit line **150**.
(56) In some possible examples, referring to FIG. **12** to FIG. **17**, removing part of the spin on hard mask layer to form a plurality of first trenches arranged at intervals and extending in a second direction (S**102**) may include the following operations.
(57) At S**1021**, an intermediate layer and a silicon-containing anti-reflection layer stacked in sequence are formed on the spin on hard mask layer, in which the silicon-containing anti-reflection layer has a plurality of first grooves arranged at intervals and extending in the second direction.
(58) Referring to FIG. **12**, an intermediate layer **300** is formed on the spin on hard mask layer **200** by a process of deposition and covers the spin on hard mask layer **200**. A silicon-containing anti-reflection (SiARC) layer **400** is formed on the intermediate layer **300** by a process of deposition and covers the intermediate layer **300**. Of course, the silicon-containing anti-reflection layer **400** may also be formed on the intermediate layer **300** by a spin on coating process. Herein, the intermediate layer **300** may be an amorphous carbon layer (ACL). The hardness of the silicon-containing anti-reflection layer **400** is relatively high, and it is not easy to collapse and deform when etching first grooves **410** of the silicon-containing anti-reflection layer **400** or other structures, and the precision of the pattern formed after the etching is good.
(59) As shown in FIG. **12**, a plurality of first grooves **410** are formed in the silicon-containing anti-reflection layer **400**, and the plurality of first grooves **410** are arranged at intervals and extend in the second direction. The first grooves **410** may penetrate through the silicon-containing anti-reflection layer **400** or may not. As shown in FIG. **12**, bottoms of the first grooves **410** are located in the silicon-containing anti-reflection layer **400**, that is, the first grooves **410** are formed in the upper part of the silicon-containing anti-reflection layer **400** far away from the base **100**, and first etching grooves are formed subsequently in the lower part of the silicon-containing anti-reflection layer **400** by utilizing the first grooves **410**. The width of the first etching grooves is less than the width of the first grooves **410**, and the density of the first etching grooves is greater than the density of the first grooves **410**, so as to further reduce the feature size of the semiconductor structure and improve the integration level of the semiconductor structure.
(60) At S**1022**, a second sacrificial layer is formed on sidewalls of the first grooves, in which the second sacrificial layer located in the first grooves defines third grooves.
(61) Specifically, referring to FIG. **12** and FIG. **13**, a second sacrificial layer **420** is deposited on sidewalls and bottoms of the first grooves **410**, and on the silicon-containing anti-reflection layer **400**. The second sacrificial layer **420** covers the surface of the side of the silicon-containing anti-reflection layer **400** facing away from the base **100**. That is, the second sacrificial layer **420** is a whole layer, which is conductive to the formation of the second sacrificial layer **420**. The material of the second sacrificial layer **420** may be silicon oxide.
(62) At S**1023**, a first filling layer is formed in the third grooves.
(63) Still referring to FIG. **12** and FIG. **13**, a first filling layer **430** is deposited in the third grooves. The material of the first filling layer **430** may be a spin on hard mask material. In some possible examples, as shown in FIG. **13**, a filling layer **430** is deposited in the third grooves and on the second sacrificial layer **420**. The first filling layer **430** is filled up the third grooves and covers the second sacrificial layer **420**.
(64) Referring to FIG. **14**, part of the first filling layer **430** and part of the second sacrificial layer **420** are removed to expose the second sacrificial layer **420** located on the sidewalls of the first grooves **410**. Specifically, the second sacrificial layer **420** located on the surface of the silicon-

containing anti-reflection layer **400** away from the base **100** and the first filling layer **430** are removed to expose the surface of the silicon-containing anti-reflection layer **400** and the second sacrificial layer **420** located on the sidewalls of the first grooves **410**. For example, a part of the first filling layer **430** and a part of the second sacrificial layer **420** are removed by a process of etching or planarization.
(65) At S**1024**, the second sacrificial layer located on the sidewalls of the first grooves is removed to form the first etching grooves.
(66) Referring to FIG. **15**, the second sacrificial layer **420** is etched to form the first etching grooves **440**. It can be understood that, in the process of etching the second sacrificial layer **420**, part of the silicon-containing anti-reflection layer **400** and part of the first filling layer **430** may be removed by the etching. The bottoms of first etching grooves **440** expose the silicon-containing anti-reflection layer **400**.
(67) It should be noted that, still referring to FIG. **15**, the selection ratio of the silicon-containing anti-reflection layer **400** and the selection ratio of the first filling layer **430** are different. After the etching is finished, there is a height difference between the silicon-containing anti-reflection layer **400** and the first filling layer **430**. Exemplarily, the etching rate of the silicon-containing anti-reflection layer **400** is slower than the etching rate of the first filling layer **430**. After the second sacrificial layer **420** is removed by etching, less height of the silicon-containing anti-reflection layer **400** is etched away during the etching, while more height of the first filling layer **430** is etched away during the etching, in the height direction, that is, in the direction perpendicular to the substrate **110**. As shown in FIG. **15**, the surface of the retained silicon-containing anti-reflection layer **400** away from the substrate **110** is higher than the surface of the retained first filling layer **430** away from the substrate **110**.
(68) At S**1025**, etching is performed along the first etching grooves to the spin on hard mask layer to form the first trenches.
(69) Referring to FIG. **16** and FIG. **17**, the silicon-containing anti-reflection layer **400**, the intermediate layer **300** and the spin on hard mask layer **200** are etched along the first etching grooves **440** to form the first trenches **210** in the spin on hard mask layer **200**. In the embodiments of the disclosure, the silicon-containing anti-reflection layer **400** is used as a transfer layer of the etching pattern to reduce the sizes of the first etching grooves **440**. In addition, no extreme ultraviolet (EUV) lithography process is used in this process, thus reducing the production cost.
(70) In some possible embodiments, referring to FIG. **3**, FIG. **4**, FIG. **18** to FIG. **20**, forming the intermediate layer and the silicon-containing anti-reflection layer stacked in sequence on the spin on hard mask layer, in which the silicon-containing anti-reflection layer has the plurality of first grooves arranged at intervals and extending in the second direction, included the following operations.
(71) (a) A first mask layer, a second mask layer and a photoresist layer stacked in sequence are formed on the silicon-containing anti-reflection layer.
(72) Referring to FIG. **3** and FIG. **4**, a first mask layer **500** is deposited on the silicon-containing anti-reflection layer **400**, in which the first mask layer **500** covers the silicon-containing anti-reflection layer **400**; a second mask layer **600** is deposited on the first mask layer **500**, in which the second mask layer **600** covers the first mask layer **500**; and a photoresist layer **700** is formed on the first mask layer **500** by a process of spin on coating, spray coating or brush coating.
(73) Exemplarily, as shown in FIG. **3** and FIG. **4**, the first mask layer **500** includes a first foundation layer **510** provided on the silicon-containing anti-reflection layer **400**, and a first anti-reflection layer **520** provided on the first foundation layer **510**; and the second mask layer **600** includes a second foundation layer **610** provided on the first anti-reflection layer **520** and a second anti-reflection layer **620** provided on the second foundation layer **610**. That is, the silicon-containing anti-reflection layer **400**, the first foundation layer **510**, the first anti-reflection layer **520**, the second foundation layer **610**, the second anti-reflection layer **620** and the photoresist layer

700 are stacked in sequence in a direction away from the base **100**.
(74) Still referring to FIG. **3** and FIG. **4**, the photoresist layer **700** is a patterned photoresist (PR) layer, that is, the photoresist layer **700** with a preset pattern is formed by a process of exposure, development and the like. The photoresist layer **700** exposes part of the second anti-reflection layer **620**. The second anti-reflection layer **620** may absorb the light used for exposuring the photoresist layer **700**, thereby reducing or preventing the reflection of the light at the second anti-reflection layer **620** to improve the accuracy of the preset pattern of the photoresist layer **700**. The material of the first foundation layer **510** is the same as the material of the second foundation layer **610**, and the material of the first anti-reflection layer **520** is the same as the material of the second anti-reflection layer **620**, so as to reduce the types of materials used during manufacturing the semiconductor structure. Exemplarily, the material of the first foundation layer **510** and the second foundation layer **610** may be a spin on hard mask composition, and the material of the first anti-reflection layer **520** and the second anti-reflection layer **620** may be silicon oxynitride.
(75) (b) The second mask layer is etched by taking the photoresist layer as a mask to form a plurality of first intermediate grooves arranged at intervals and extending in the second direction in the second mask layer.
(76) Referring to FIG. **4** and FIG. **18**, the second mask layer **600** is etched by taking the photoresist layer **700** as a mask. The second mask layer **600** not covered by the photoresist layer **700** is removed, meanwhile the second mask layer **600** covered by the photoresist layer **700** is retained. A plurality of first intermediate grooves **630** arranged at intervals and extending in the second direction are formed in the second mask layer **600**. The first intermediate grooves **630** penetrate through the second mask layer **600** to expose the first mask layer **500**. In this way, the preset pattern of the photoresist layer **700** is transferred to the second mask layer **600**, and the first intermediate grooves **630** are formed in the second mask layer **600**.
(77) (c) A third sacrificial layer is deposited on sidewalls and bottoms of the first intermediate grooves, and on the second mask layer. The third sacrificial layer located in the first intermediate grooves defines second intermediate grooves.
(78) Referring to FIG. **18** and FIG. **19**, a third sacrificial layer **640** is deposited on sidewalls and bottoms of the first intermediate grooves **630**, and on the second mask layer **600**. For example, the third sacrificial layer **640** is formed by a process of atomic layer deposition process, so as to form the third sacrificial layer **640** with a better quality. The material of the third sacrificial layer **640** may be silicon oxide.
(79) (d) The third sacrificial layer located on a top of the second mask layer and bottoms of the second intermediate grooves is removed to retain the third sacrificial layer located on the sidewalls of the first intermediate grooves.
(80) Referring to FIG. **20**, the third sacrificial layer **640** located on a top of the second mask layer **600** and bottoms of the second intermediate grooves **650** is removed by a process of etching to retain the third sacrificial layer **640** located on the sidewalls of the first intermediate grooves **630**. After the etching, the second mask layer **600** and the first mask layer **500** are exposed. That is, the third sacrificial layer **640** located on the sidewalls of the first intermediate grooves **630** is formed by depositing and then back etching.
(81) (e) The second mask layer, the first mask layer and the silicon-containing anti-reflection layer are etched by taking the retained third sacrificial layer as a mask to form the first grooves.
(82) Referring to FIG. **20** and FIG. **12**, the second mask layer **600** sandwiched by the third sacrificial layer **640**, the first mask layer **500** and the silicon-containing anti-reflection layer **400** are removed by etching to form the first grooves **410** in the silicon-containing anti-reflection layer **400**. After the first grooves **410** are formed, other layers on the silicon-containing anti-reflection layer **400** are removed to expose the silicon-containing anti-reflection layer **400**.
(83) It should be noted that, the operations of forming the intermediate layers and the silicon-containing anti-reflection layer stacked in sequence on the spin on hard mask layer, in which the

silicon-containing anti-reflection layer has a plurality of first grooves arranged at intervals and extending in the second direction, can also be performed in other ways. In other possible examples, it includes the following operations.
(84) (a′) A first mask layer, a second mask layer and a photoresist layer stacked in sequence are formed on the silicon-containing anti-reflection layer.
(85) (b′) The second mask layer is etched by taking the photoresist layer as a mask to form a plurality of first intermediate grooves arranged at intervals and extending in the second direction in the second mask layer.
(86) (c′) A third sacrificial layer is deposited on sidewalls and bottoms of the first intermediate grooves, and on the second mask layer. The third sacrificial layer located in the first intermediate grooves defines second intermediate grooves.
(87) The operations of (a′), (b′) and (c′) in the example can refer to the operations of (a), (b) and (c) in the above example, and will not be repeated here.
(88) (d′) A second filling layer is formed in the second intermediate grooves and on the third sacrificial layer.
(89) The second filling layer is formed by a process of deposition. The second filling layer fills up the second intermediate grooves and covers the third sacrificial layer. The material of the second filling layer may be a spin on hard mask composition.
(90) (e′) Part of the second filling layer and part of the third sacrificial layer are removed to expose the third sacrificial layer located on the sidewalls of the first intermediate grooves.
(91) Exemplarily, part of the second filling layer located on the surface of the second mask layer away from the base and part of the third sacrificial layer are removed by a process of planarization to expose the surface as well as the third sacrificial layer located on the sidewalls of the first intermediate grooves.
(92) (f) The third sacrificial layer, the first mask layer and the silicon-containing anti-reflection layer are etched to form the first grooves.
(93) The third sacrificial layer, the first mask layer and the silicon-containing anti-reflection layer below the third sacrificial layer are etched to form the first grooves in the silicon-containing anti-reflection layer. During the etching, layers above the silicon-containing anti-reflection layer are also at least partially removed. The retained layer can be removed alone by a process of etching, and after the retained layer is removed, the silicon-containing anti-reflection layer is exposed.
(94) It should be noted that, in the different examples of forming the intermediate layer and the silicon-containing anti-reflection layer arranged in sequence on the spin on hard mask layer and the silicon-containing anti-reflection layer having the plurality of first grooves arranged at intervals and extending in the second direction, the patterns of the photoresist layers are different to ensure that the positions of the first grooves finally formed are the same.
(95) In the embodiments of the disclosure, the first grooves **410** are formed in the silicon-containing anti-reflection layer **400** by a process of self-aligned double patterning (SADP), and the feature size of the formed first grooves **410** is reduced, while the density of them is increased. In addition, in the subsequent process of forming the first etching grooves **440** in the silicon-containing anti-reflection layer **400**, a process of self-aligned double patterning is also performed, so that the feature size of the first etching grooves **440** is reduced and the density of them is increased, thereby further improving the integration level of the semiconductor structure formed subsequently.
(96) In a possible example of the disclosure, referring to FIG. **12** to FIG. **14**, a second sacrificial layer is deposited on sidewalls and bottoms of the first grooves, and on the silicon-containing anti-reflection layer. Accordingly, forming a first filling layer in the third grooves includes the following operations.
(97) Referring to FIG. **13**, a filling layer **430** is deposited in the third grooves and on the second sacrificial layer **420**. As shown in FIG. **13**, the first filling layer **430** is filled up the third grooves

and covers the second sacrificial layer **420**, that is, the surface of the first filling layer **430** away from the base **100** is higher than the surface of the second sacrificial layer **420** away from the base **100**.
(98) Referring to FIG. **14**, part of the first filling layer **430** and part of the second sacrificial layer **420** are removed to expose the second sacrificial layer **420** located on the sidewalls of the first grooves **410**. As shown in FIG. **14**, the part of the first filling layer **430** and the part of the second sacrificial layer **420** are removed by a process of planarization to expose the silicon-containing anti-reflection layer **400** and the second sacrificial layer **420**.
(99) In the embodiments of the disclosure, after the first trenches are formed, forming first sacrificial layers in the first trenches includes the following operation. The first sacrificial layers **220** are deposited in the first etching grooves and the first trenches **210**. Referring to FIG. **21**, the first sacrificial layers **220** are filled in the first trenches **210** and the first etching grooves, and may cover the intermediate layer **300** or the silicon-containing anti-reflection layer **400**. It can be understood that during the etching of the spin on hard mask layer **200** to form the first trenches **210**, part of the silicon-containing anti-reflection layer **400** is also etched away. The first sacrificial layers **220** cover the silicon-containing anti-reflection layer **400** when the silicon-containing anti-reflection layer **400** is not completely removed. The first sacrificial layers **220** cover the intermediate layer **300**, when the silicon-containing anti-reflection layer **400** is removed completely and the intermediate layer **300** is exposed.
(100) It should be noted that, after the first sacrificial layers are formed in the first trenches, the method further includes the following operation. The intermediate layer, the first sacrificial layer, and the silicon-containing anti-reflection layer located on the spin on hard mask layer are removed to expose the spin on hard mask layer. Referring to FIG. **22**, other layers located on the spin on hard mask layer **200** are removed by a process of planarization to expose the spin on hard mask layer **200**, in order to conveniently remove the spin on hard mask layer **200** subsequently.
(101) In some possible examples of the disclosure, after the first sacrificial layers are removed to expose the first trenches and elongate the filling holes of the first trenches to the active areas, the method further includes the following operation. Conductive pillars are formed in the filling holes, in which the conductive pillars are respectively electrically connected to the active areas.
(102) Referring to FIG. **10**, FIG. **23** to FIG. **25**, each conductive pillar **250** is in contact with an active area **111** to achieve the electrical connection between the conductive pillar **250** and the active area **111**. Each conductive pillar **250** is provided in each filling hole **212**. The conductive pillars **250** are not connected with each other. Active areas **111** and word lines **113** are provided in the substrate **110**, and the word lines **113** are insulated from the active areas **111** and staggered from the conductive pillars **250**. Exemplarily, the word lines **113** pass through the middle areas of the active areas **111**, and the conductive pillars **250** are electrically connected to the end areas of the active areas **111**, respectively.
(103) Specifically, referring to FIG. **11**, FIG. **23** and FIG. **24**, forming conductive pillars **250** in the filling holes **212**, in which the conductive pillars **250** are respectively electrically connected to the active areas **111**, includes the following operations.
(104) A first conductive layer **251** is deposited in the first trenches **210** and on the first supporting layers **240**, in which the first conductive layer **251** is filled in the first trenches **210** and covers the first supporting layers **240**, and the first conductive layer **251** is electrically connected to the active areas **111**. As shown in FIG. **11** and FIG. **23**, a first conductive layer **251** is filled up the first trenches **210** and covers the first supporting layers **240**. The material of the first conductive layer **251** may be silicon oxide.
(105) After the first conductive layer **251** is formed, the first conductive layer **251** is etched to retain the part of the first conductive layer **251** located in the filling holes **212**, in which the retained first conductive layer **251** forms the conductive pillars **250**. As shown in FIG. **11** and FIG. **24**, the first conductive layer **251** located on the bit lines and the filling holes **212** is removed to

retain the first conductive layer **251** located at the bottoms of the filling holes **212**. The retained first conductive layer **251** is separated into a plurality of conductive layers by the first supporting layers **240** and the bit lines. The conductive pillar **250** are arranged at intervals and not connected with each other. Specifically, as shown in FIG. **24**, at least the part of the first conductive layer **251** located on second supporting layers **152** of the bit lines is removed to separate the first conductive layer **251** into a plurality of conductive layers.

(106) It should be noted that, after the conductive pillars **250** are formed in the filling holes **212**, in which the conductive pillars **250** are electrically connected to the active areas **111**, the method further include the following operation. A protective layer **260** is deposited on each of the conductive pillars **250**, in which the protective layer **260** covers the conductive pillar **250**. Referring to FIG. **25**, a protective layer **260** is deposited on each of the conductive pillars **250**, in which the protective layer **260** covers the surface of the conductive pillar **250** away from the base **100**. The material of the protective layer **260** may be a nitride, for example, silicon nitride. The thickness of the protective layer **260** is relatively small. For example, the protective layer **260** can reduce or prevent the conductive pillar **250** from being oxidized by being exposed to air.

(107) The embodiments of the disclosure also provide a semiconductor structure, which is obtained by the above-mentioned method, and thus at least has the advantage that the loss of a bit line **150** is less. Specific effects are described above and will not be repeated here.

(108) In this specification, examples or embodiments are described in a progressive way, and each embodiment focuses on the differences with other embodiments, and the same and similar parts of the embodiments can be referred to each other.

(109) In the description of this specification, descriptions of the reference terms “an embodiment”, “some embodiments”, “exemplary embodiment”, “example”, “specific example”, or “some examples”, etc. mean that specific features, structures, materials, or features described with reference to the embodiment or example are included in at least one embodiment or example of the disclosure. In the specification, the illustrative use of the above terms does not necessarily refer to the same embodiment or example. Furthermore, the specific features, structures, materials or characteristics described may be combined in any one or more embodiments or examples in a proper manner.

(110) Finally, it should be noted that, the above embodiments are only used to illustrate the technical solution of the disclosure, but not to limit the disclosure. Although the disclosure has been described in detail with reference to the foregoing embodiments, an ordinary person skilled in the art should understand that it is still possible to modify the technical solution described in the foregoing embodiments or to replace some or all of their technical features equivalently. However, these modifications or replacements do not make the essence of corresponding technical solution depart from the scope of the technical solution of the embodiments of the disclosure.

Claims

1. A method for manufacturing a semiconductor structure, comprising: forming a spin on hard mask layer on a base, wherein a plurality of active areas arranged at intervals are provided in the base, a plurality of bit lines arranged at intervals and extending in a first direction are provided on the base, each of the bit lines is electrically connected to at least one of the active areas, and the spin on hard mask layer is filled between the bit lines and covers the bit lines; removing part of the spin on hard mask layer to form a plurality of first trenches arranged at intervals and extending in a second direction; forming first sacrificial layers in the first trenches, wherein each of the first sacrificial layers is filled in each of the first trenches; removing the spin on hard mask layer between the first sacrificial layers to form second trenches; forming first supporting layers in the second trenches, wherein each of the first supporting layers is filled in each of the second trenches; and removing the first sacrificial layers and elongating the first trenches located between adjacent

bit lines to the active areas.
2. The method according to claim 1, wherein removing part of the spin on hard mask layer to form the plurality of first trenches arranged at intervals, wherein the first trenches expose the base, comprises: forming an intermediate layer and a silicon-containing anti-reflection layer stacked in sequence on the spin on hard mask layer, wherein the silicon-containing anti-reflection layer has a plurality of first grooves arranged at intervals and extending in the second direction; forming a second sacrificial layer on sidewalls of the first grooves, wherein the second sacrificial layer located in the first grooves defines third grooves; forming a first filling layer in the third grooves; removing the second sacrificial layer located on the sidewalls of the first grooves to form first etching grooves; and etching along the first etching grooves to the spin on hard mask layer to form the first trenches.
3. The method according to claim 2, wherein bottoms of first grooves are located in the silicon-containing anti-reflection layer.
4. The method according to claim 2, wherein forming the intermediate layer and the silicon-containing anti-reflection layer stacked in sequence on the spin on hard mask layer, wherein the silicon-containing anti-reflection layer has the plurality of first grooves arranged at intervals and extending in the second direction, comprises: forming a first mask layer, a second mask layer and a photoresist layer stacked in sequence on the silicon-containing anti-reflection layer; etching the second mask layer by taking the photoresist layer as a mask to form a plurality of first intermediate grooves arranged at intervals and extending in the second direction in the second mask layer; depositing a third sacrificial layer on sidewalls and bottoms of the first intermediate grooves, and on the second mask layer, wherein the third sacrificial layer located in the first intermediate grooves defines second intermediate grooves; removing the third sacrificial layer located on a top of the second mask layer and bottoms of the second intermediate grooves to retain the third sacrificial layer located on the sidewalls of the first intermediate grooves; and etching the second mask layer, the first mask layer and the silicon-containing anti-reflection layer by taking the retained third sacrificial layer as a mask to form the first grooves.
5. The method according to claim 4, wherein the first mask layer comprises a first foundation layer provided on the silicon-containing anti-reflection layer, and a first anti-reflection layer provided on the first foundation layer; the second mask layer comprises a second foundation layer provided on the first anti-reflection layer and a second anti-reflection layer provided on the second foundation layer; and a material of the first foundation layer is same as a material of the second foundation layer, and a material of the first anti-reflection layer is same as a material of the second anti-reflection layer.
6. The method according to claim 2, wherein forming the second sacrificial layer on the sidewalls of the first grooves, wherein the second sacrificial layer located in the first grooves defines the third grooves, comprises: depositing the second sacrificial layer on the sidewalls and bottoms of the first grooves and on the silicon-containing anti-reflection layer.
7. The method according to claim 6, wherein forming the first filling layer in the third grooves comprises: forming the first filling layer in the third grooves and on the second sacrificial layer; and removing part of the first filling layer and part of the second sacrificial layer to expose the second sacrificial layer located on the sidewalls of the first grooves.
8. The method according to claim 7, wherein forming the first sacrificial layers in the first trenches comprises: depositing the first sacrificial layers in the first etching grooves and the first trenches.
9. The method according to claim 8, further comprising: after forming the first sacrificial layers in the first trenches, removing the intermediate layer, the first sacrificial layers, and the silicon-containing anti-reflection layer located on the spin on hard mask layer to expose the spin on hard mask layer.
10. The method according to claim 1, further comprising: after removing the first sacrificial layers to expose the first trenches and elongating filling holes of the first trenches to the active areas,

forming conductive pillars in the filling holes, wherein the conductive pillars are electrically connected to the active areas.

11. The method according to claim 1, wherein the base comprises a substrate, an insulating layer disposed on the substrate, and a barrier layer disposed on the insulating layer, wherein the active areas are disposed in the substrate, and the insulating layer covers the active areas, and wherein removing the first sacrificial layers and elongating the first trenches located between adjacent bit lines to the active areas comprises: etching the first sacrificial layers to expose the first trenches, wherein each of the first trenches comprises a second groove located on the bit lines and extending in the second direction, and filling holes located between adjacent bit lines and in communication with the second groove; and etching the insulating layer along the filling holes of the first trenches to expose the active areas by the filling holes.

12. The method according to claim 10, wherein forming conductive pillars in the filling holes, wherein the conductive pillars are electrically connected to the active areas, comprises: depositing a first conductive layer in the first trenches and on the first supporting layers, wherein the first conductive layer is filled in the first trenches and covers the first supporting layers, and the first conductive layer is electrically connected to the active areas; and etching the first conductive layer to retain part of the first conductive layer located in the filling holes, wherein the retained first conductive layer forms the conductive pillars.

13. The method according to claim 12, wherein a plurality of word lines arranged at intervals and extending in the second direction are further disposed in a substrate, and the word lines are insulated from the active areas and staggered from the conductive pillars.

14. The method according to claim 10, further comprising: after forming conductive pillars in filling holes, wherein the conductive pillars are electrically connected to the active areas, depositing a protective layer on each of the conductive pillars, wherein the protective layer covers the conductive pillars.

15. The method according to claim 1, wherein each of the bit lines comprises a second conductive layer and a second supporting layer covering the second conductive layer, in the second supporting layer, an oxide layer is provided at a side of the second conductive layer, and a material of the second supporting layer and a material of the first supporting layer are the same, wherein a selection ratio between the spin on hard mask layer and the second supporting layer is greater than or equal to 5.

16. The semiconductor structure obtained by the method for manufacturing the semiconductor structure according to claim 1.